Checking for Prior-Data Conflict

Evans & Moshonov (2006)

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Introduction: The Bayesian Framework

Bayesian inference is built on the combination of two fundamental components:

- **The Sampling Model**: $f(x \mid \theta)$, representing the process generating the data.
- **The Prior Distribution**: $\pi(\theta)$, representing our beliefs about θ before seeing data.

The joint distribution is $P(X, \theta) = f(x \mid \theta)\pi(\theta)$.

The Challenge

If either the sampling model or the prior is "wrong," the resulting posterior inference $\pi(\theta \mid x)$ can be highly misleading.

Two Types of Model Failure

Evans and Moshonov distinguish between two ways a Bayesian model can fail:

1. **Sampling Model Failure**: The observed data x is unlikely for *any* value of θ that is reasonable. This is standard "model checking."
2. **Prior-Data Conflict**: The observed data x is unlikely for the values of θ where the prior $\pi(\theta)$ places most of its mass.

The Focus

This paper focuses specifically on detecting the second type: **Prior-Data Conflict**.

Defining Prior-Data Conflict

Conceptual Definition

A prior-data conflict exists when the observed data x falls in the tails (low-density region) of the **prior predictive distribution**:

$$m(x) = \int_{\Theta} f(x \mid \theta)\pi(\theta)\, d\theta$$

The p-value approach: We calculate the probability of observing data "as or more surprising" than x_{obs}:

$$p = P(m(X) \leq m(x_{obs}))$$

A small p-value indicates that the data and the prior are in conflict.

The Role of Sufficient Statistics

The paper argues that we shouldn't check the entire data vector x, but rather a **minimal sufficient statistic** $T(x)$.

- **Theorem**: If T is sufficient, then $P(X \mid T)$ is independent of θ and the prior.
- **Conclusion**: All information about the prior-data conflict is contained in the marginal distribution of T, $m_T(t)$.

Checking T instead of X isolates the prior conflict from general model failures.

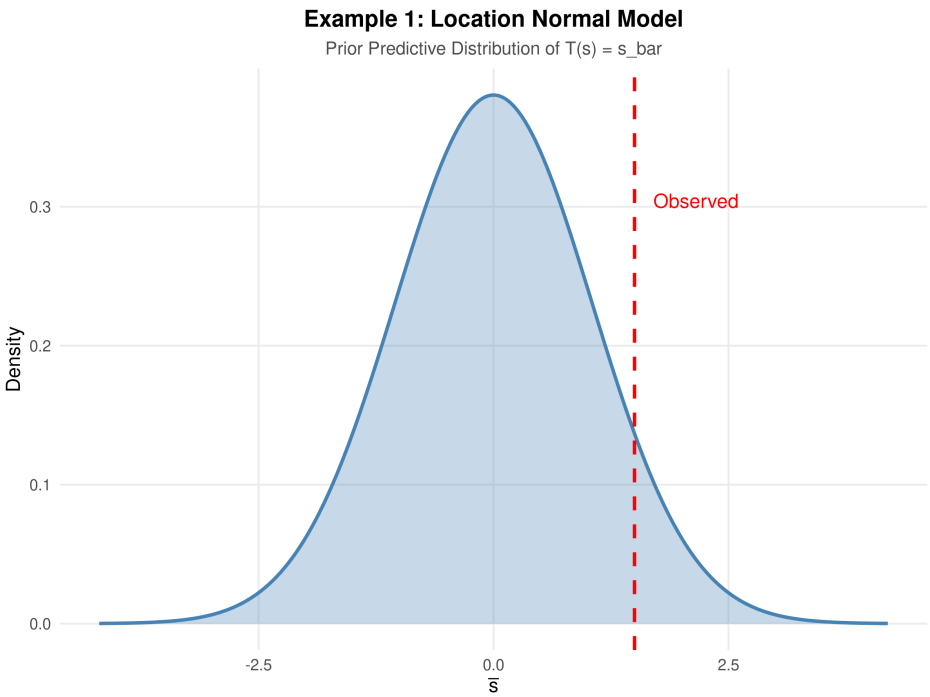
Example 1: Location Normal Model

Scenario: Estimating the mean θ of a population with known variance $\sigma^2 = 1$.

- **Model**: $X_1, \ldots, X_n \sim N(\theta, 1)$.
- **Prior**: $\theta \sim N(\theta_0, \sigma_0^2)$.
- **Sufficient Statistic**: The sample mean $\bar{X}$.
- **Predictive**: $\bar{X} \sim N(\theta_0, \sigma_0^2 + 1/n)$.
- **p-value**:

$$P(m_T(\bar{s}) \leq m_T(\bar{s}_0)) = 2(1 - \Phi(|\bar{s}_0 - \theta_0|/(\sigma_0^2 + 1/n)^{1/2})).$$

Example 1: Location Normal Model



Key Insights:

- The plot shows the density of $\bar{X}$ under the prior predictive.
- The red dot marks the observed $\bar{x}$.
- If the red dot is in the tails, p is small, indicating conflict.

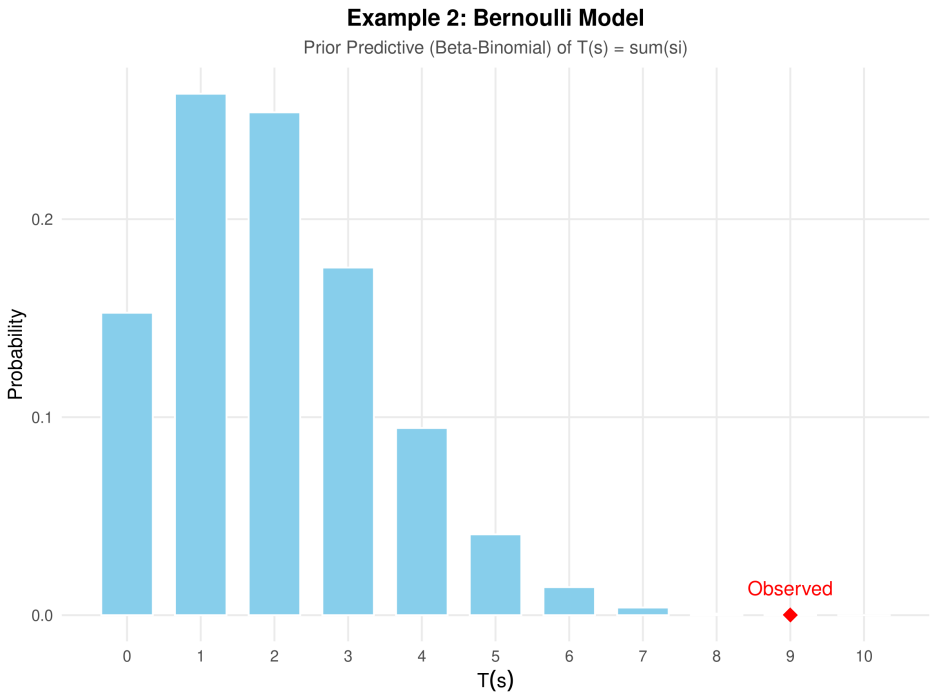
Example 2: Bernoulli Model

Scenario: Estimating the probability of success θ in n trials.

- **Model**: $X_i \sim \text{Bernoulli}(\theta)$.
- **Prior**: $\theta \sim \text{Beta}(\alpha, \beta)$.
- **Sufficient Statistic**: Total number of successes $S = \sum X_i$.
- **Predictive**: Beta-Binomial distribution.
- **p-value**:

$$P\left(\frac{\Gamma(t+\alpha)\Gamma(n-t-\beta)}{\Gamma(t+1)\Gamma(n-t+1)} \leq \frac{\Gamma(t_0+\alpha)\Gamma(n-t_0+\beta)}{\Gamma(t+1)\Gamma(n-t_0+1)}\right).$$

Example 2: Bernoulli Model



Key Insights:

- Discrete case: p is the sum of probabilities for all s where $m(s) \leq m(s_{obs})$.
- Conflict occurs if s_{obs} is very different from the prior expectation $\frac{n\alpha}{\alpha+\beta}$.

Example 4: Location-Scale Normal

Scenario: Both μ and σ^2 are unknown parameters.

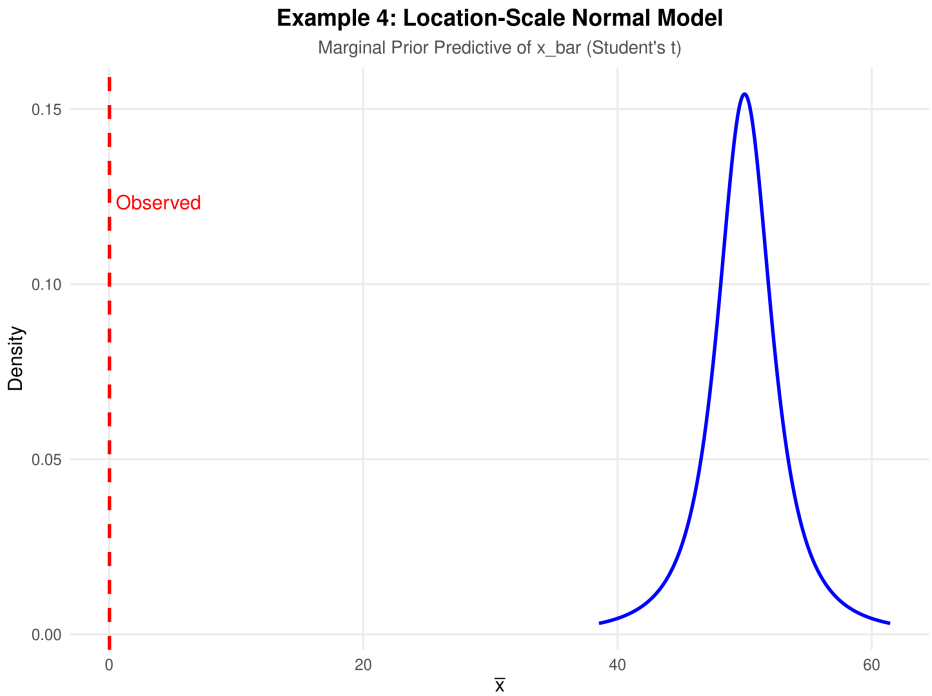
- **Model**: $X_i \sim N(\mu, \sigma^2)$.
- **Prior**: $\sigma^2 \sim \text{IG}(\alpha_0, \beta_0)$ and $\mu \mid \sigma^2 \sim N(\mu_0, \tau_0^2\sigma^2)$.
- **Sufficient Statistics**: $(\bar{X}, S^2)$.
- **Approach**: Check the joint consistency of location and scale.
- **Predictive**: Is proportional to

$$(S^2)^{(n-1)/2-1}\beta_x^{-n/2-\alpha_0}$$

where

$$\beta_x = \beta_0 + (n-1)S^2/2 + n(\bar{X} - \mu_0)/2(n\tau_0^2 + 1).$$

Example 4: Location-Scale Normal



Key Insights:

- The plot represents the joint prior predictive density.
- Conflict can arise from an unusual mean, an unusual variance, or both.

The Problem of Ancillarity

A statistic $U(X)$ is **ancillary** if its distribution does not depend on θ.

The Issue

Sometimes $T(X)$ contains variation that is purely ancillary. If we don't account for this, we might mistake "unusual data shape" for "prior conflict."

The Solution: Condition the conflict check on the observed value of the ancillary statistic:

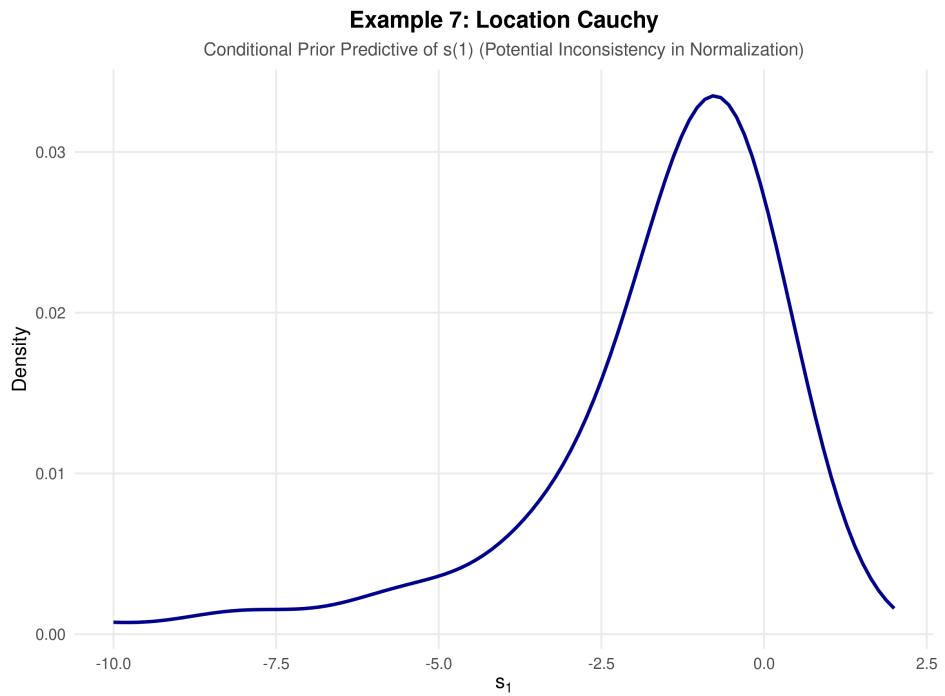
$$p = P(m_T(T \mid U = u_{obs}) \leq m_T(t_{obs} \mid U = u_{obs}))$$

This "filters out" variation that the prior cannot possibly explain.

Example 7: Location Cauchy

$X_i \sim \text{Cauchy}(\theta, 1)$. This is a classic case where the "shape" of the sample (distances between points) is ancillary.

- We condition on the ancillary "shape" to check the "location" conflict.



Note: Conditioning ensures we only penalize the prior for missing the **center** of the data, not for the data having "heavy tails."

Section 3: Noninformative Priors

How do noninformative (diffuse) priors behave under this check?

Intuition

A noninformative prior is one that is "spread out." Therefore, it should be hard to find data that it *doesn't* support.

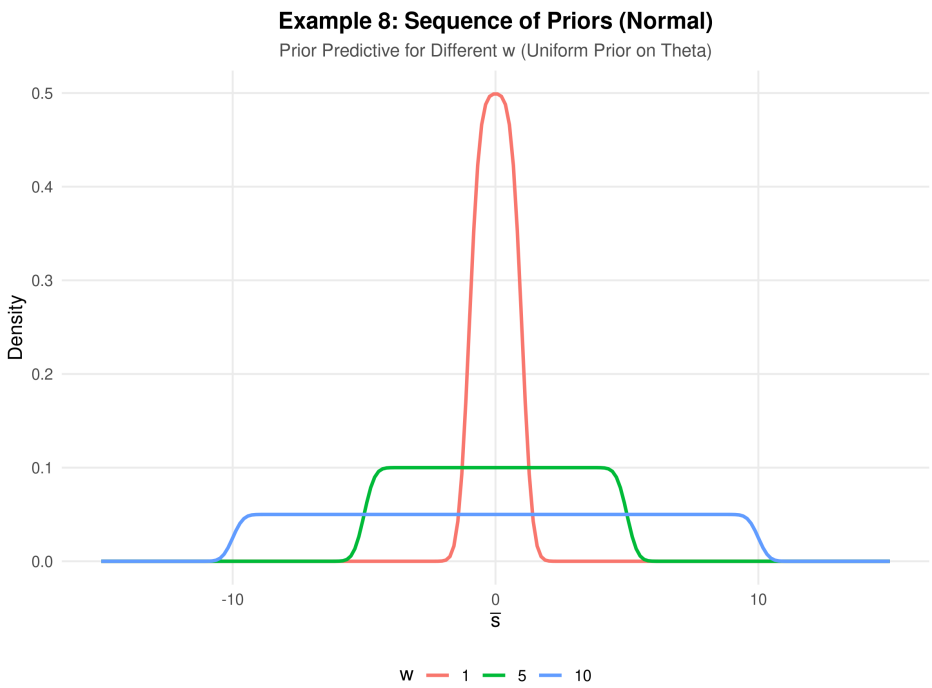
The Property: As the prior becomes more diffuse (variance $\to \infty$), the p-value for conflict should go to 1.

Example 8: Sequence of Normal Priors

Scenario: Testing the sensitivity of the conflict check to prior uncertainty.

- **Prior**: $\theta \sim N(0, \tau^2)$ for increasing values of τ^2.
- **Theory**: As $\tau^2 \to \infty$, the prior becomes noninformative.

Example 8: Sequence of Normal Priors



Key Insights:

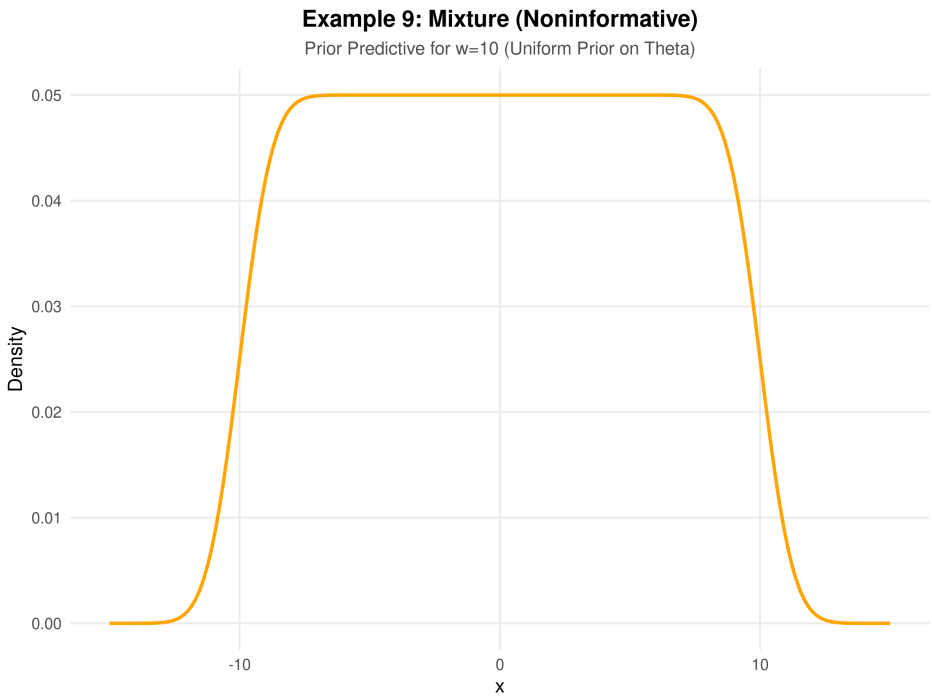
- The plot shows that as the prior becomes more diffuse, the predictive density covers more ground, and conflict disappears.

Example 9: Uniform Prior Sensitivity

Scenario: Using a flat prior on a bounded interval $[-w, w]$.

- **Prior**: $\theta \sim \text{Uniform}(-w, w)$.
- **Theory**: Larger w implies less prior information.

Example 9: Uniform Prior Sensitivity



Key Insights:

- Demonstrates that noninformative priors are "safe" from conflict by design.

Practical vs. Statistical Significance

Even if a conflict is statistically significant ($p < 0.05$), does it actually **matter** for our final inference?

- If we have a huge amount of data, the likelihood dominates, and a "bad" prior might not change the posterior much.

The Effect Size

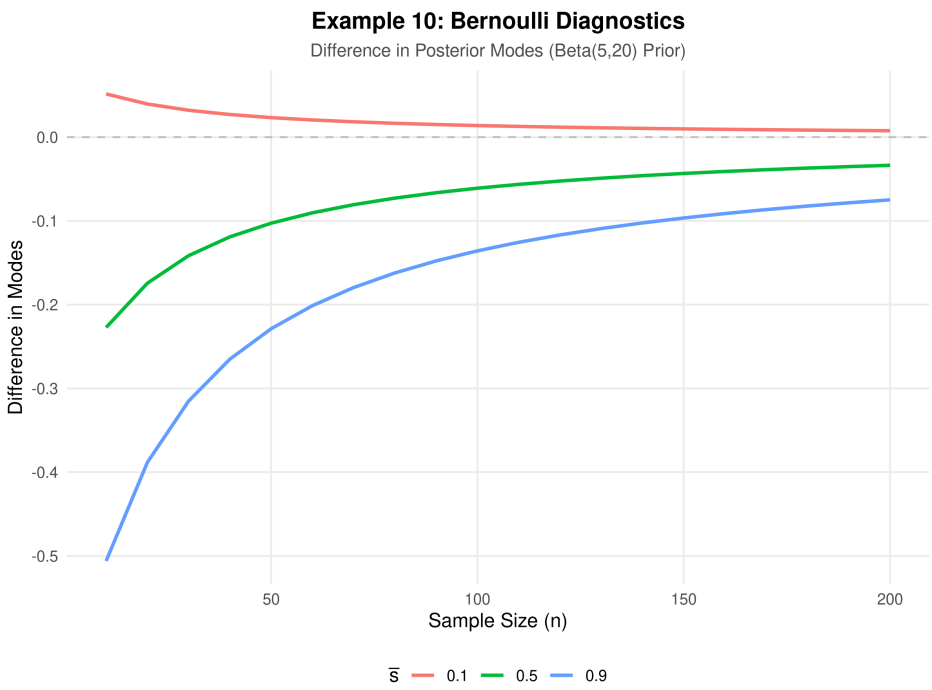
Compare the posterior mean using the informative prior (π) vs. a noninformative prior (π_*).

Example 10: Bernoulli Practical Significance

Scenario: Suppose scenario of Example 2. Does a conflict matter if the sample size n is large?

- **Method**: Compare the posterior mode under the informative prior vs. a flat prior.

Example 10: Bernoulli Practical Significance



Key Insights:

- As n increases, the difference (bias) decreases, even if the p-value still indicates conflict.

Section 5: Hierarchical Models

In hierarchical models, conflict can happen at different levels:

1. **Data level**: Data vs. the entire hierarchy.
2. **Parameter level**: Group parameters vs. the hyper-prior.

The paper suggests a sequential checking approach.

The Factorization of the Joint Distribution

The paper's most significant theoretical contribution is the factorization:

$$P(X, \theta) = \underbrace{P(X \mid T)}_{\text{Model Checking}} \times \underbrace{m_T(T)}_{\text{Prior Conflict}} \times \underbrace{\pi(\theta \mid T)}_{\text{Inference}}$$

- **Part 1**: Checks if the sampling model is fundamentally flawed.
- **Part 2**: Checks if the prior is consistent with the data.
- **Part 3**: Performs the actual Bayesian inference.

Why This Factorization Matters

- It shows that model checking and prior checking are **orthogonal** tasks.
- You can have a perfect sampling model but a terrible prior.
- You can have a perfect prior for a model that doesn't fit the data.

Methodological Rule

Check the sampling model first. If it fails, the prior check is meaningless.

Summary of the Evans-Moshonov Approach

1. Identify minimal sufficient statistics T.
2. Derive the prior predictive $m_T(t)$.
3. Identify and condition on ancillary statistics U.
4. Calculate the p-value for conflict.
5. If conflict exists, assess its practical significance (Effect Size).

Discussion: What to do when conflict is found?

If you detect a prior-data conflict, you have several options:

- **Broaden the prior**: Increase the variance (make it more noninformative).
- **Re-examine the source**: Was the prior based on expert opinion that might be biased or outdated?
- **Report it**: Even if you don't change the prior, the conflict is a key part of the scientific result.

Conclusion

- Prior-data conflict is a distinct and detectable form of model failure.
- "Subjective" priors are powerful but risky.
- Ancillarity is crucial for "fair" conflict checking.
- This paper moves Bayesian analysis toward a more "critical" and "self-validating" workflow.

Reference

- Evans, M., and Moshonov, H. (2006). "Checking for Prior-Data Conflict." *Bayesian Analysis*, 1(4), 893-914.